General Arthrology

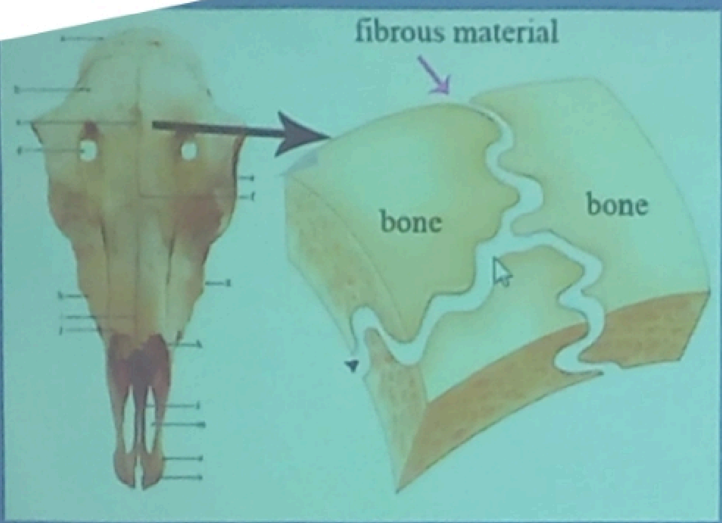
- **Def**. It is the science dealing with the study of joint:

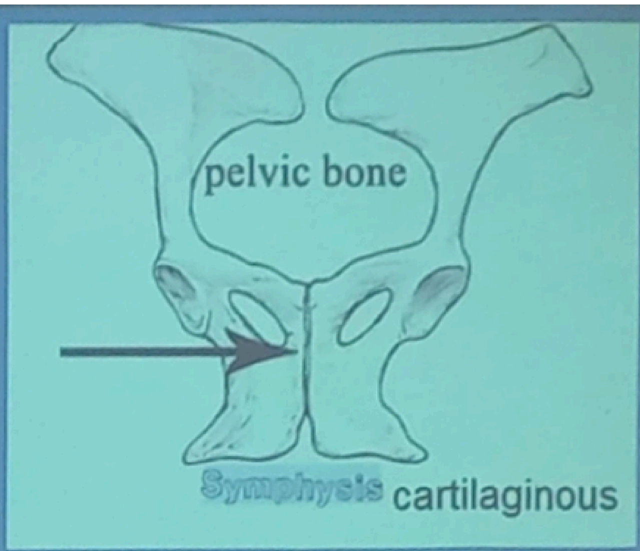
Joint (articulation):

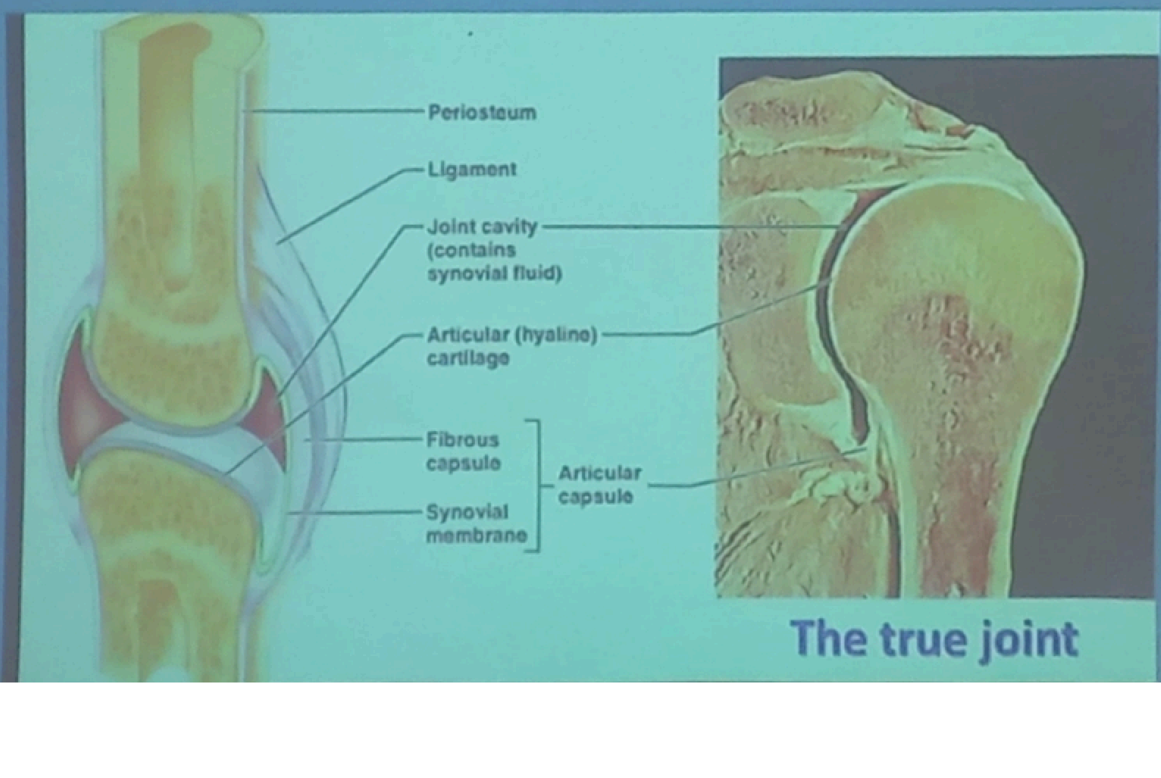
- Is formed by union of two or more bones or cartilages by other tissue (uniting medium)
- Fibrous tissue- Cartilage- synovial fluid

Classification of the joint according to the uniting medium:

- Fibrous joint (synarthrosis)
- Cartilaginous joint (amphiarthrosis).
- Synovial joint (diarthrosis).

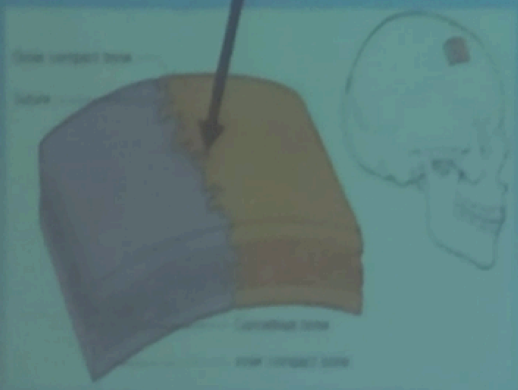

fibrous material
bone
bone


pelvic bone
Symphysis cartilaginous

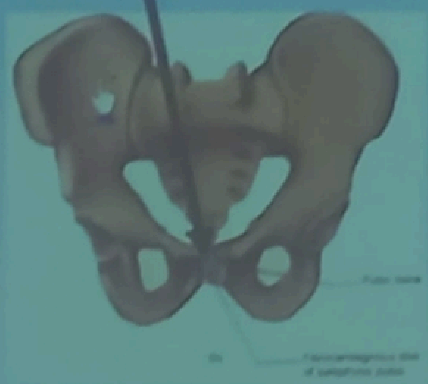

Periosteum
Ligament
Joint cavity (contains synovial fluid)
Articular (hyaline) cartilage
Fibrous capsule
Synovial membrane
Articular capsule
The true joint

Classification of joints:

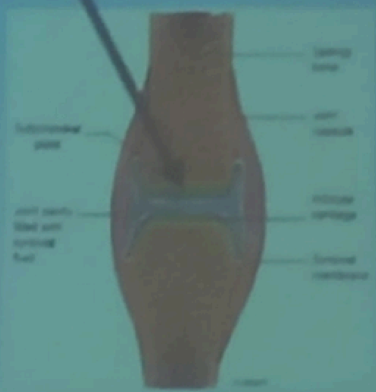
Fibrous joints



Cartilaginous joints



Synovial joints.



Fibrous joints

This type is characterized by:

- The two bones or more are united by fibrous tissue.
- No joint cavity.
- immovable joint
- Most of these joints are temporary

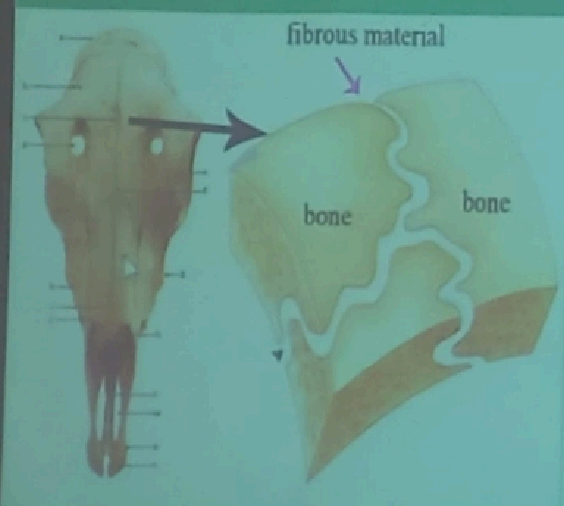
Fibrous tissue → become bone (ossification) → synostosis (immovable joint)

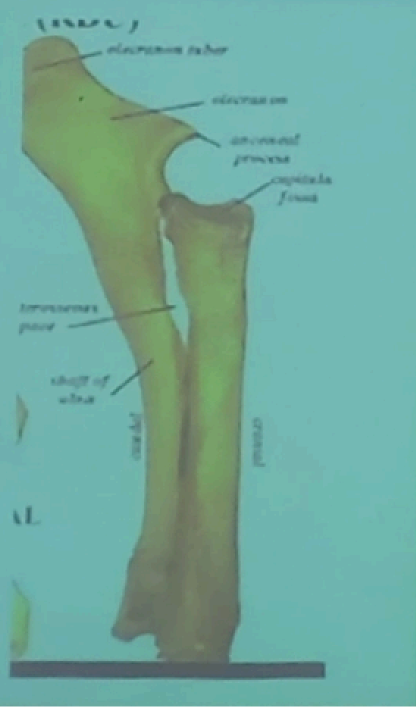
Classification of fibrous joint:

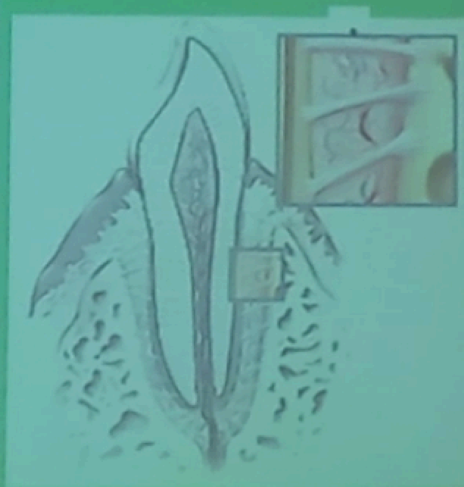
1- suture (skull) 2- **Syndesmosis (outsid the skull)** **3-Gomphosis(teeth)**

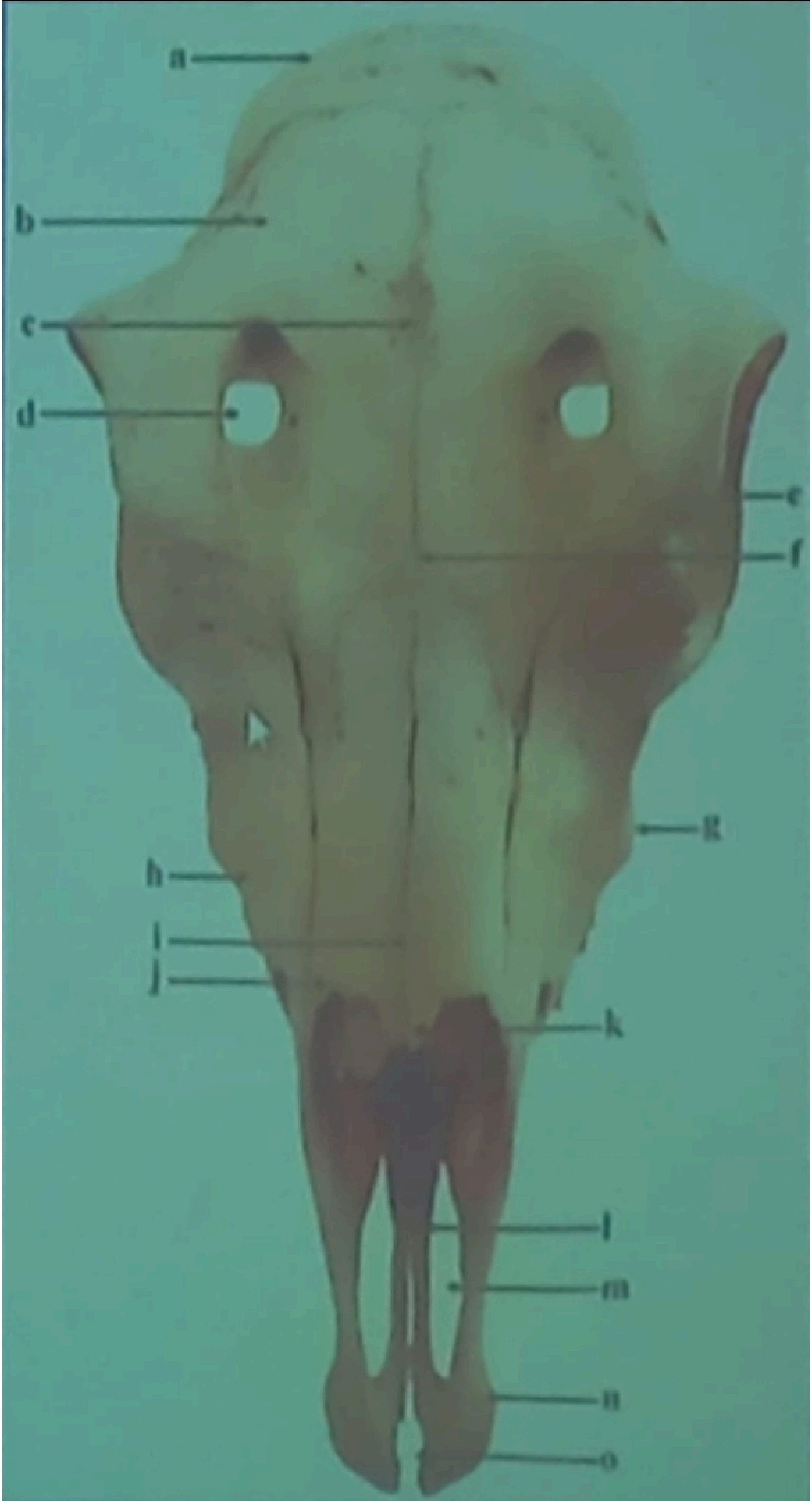
a Abdelqader

suture	Syndesmosis	Gomphosis
The fibrous joints in the skull	fibrous joint outside the skull. Ex. union of shafts of Rdius ulna	implantation of the teeth in the alveoli by Fibrous Periodontal Ligaments





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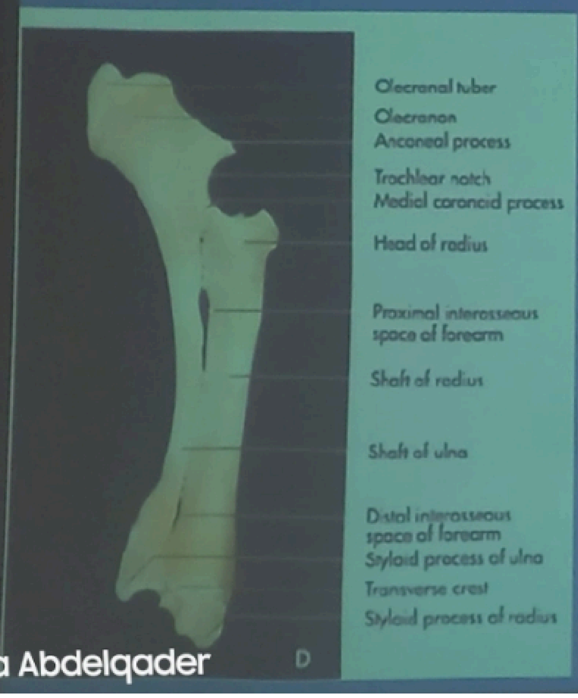
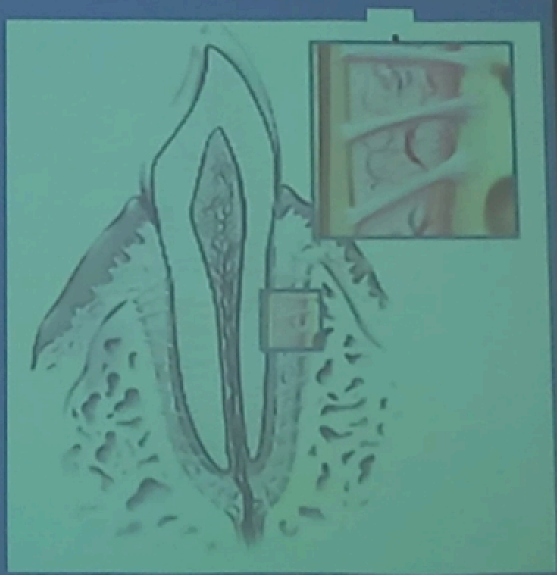
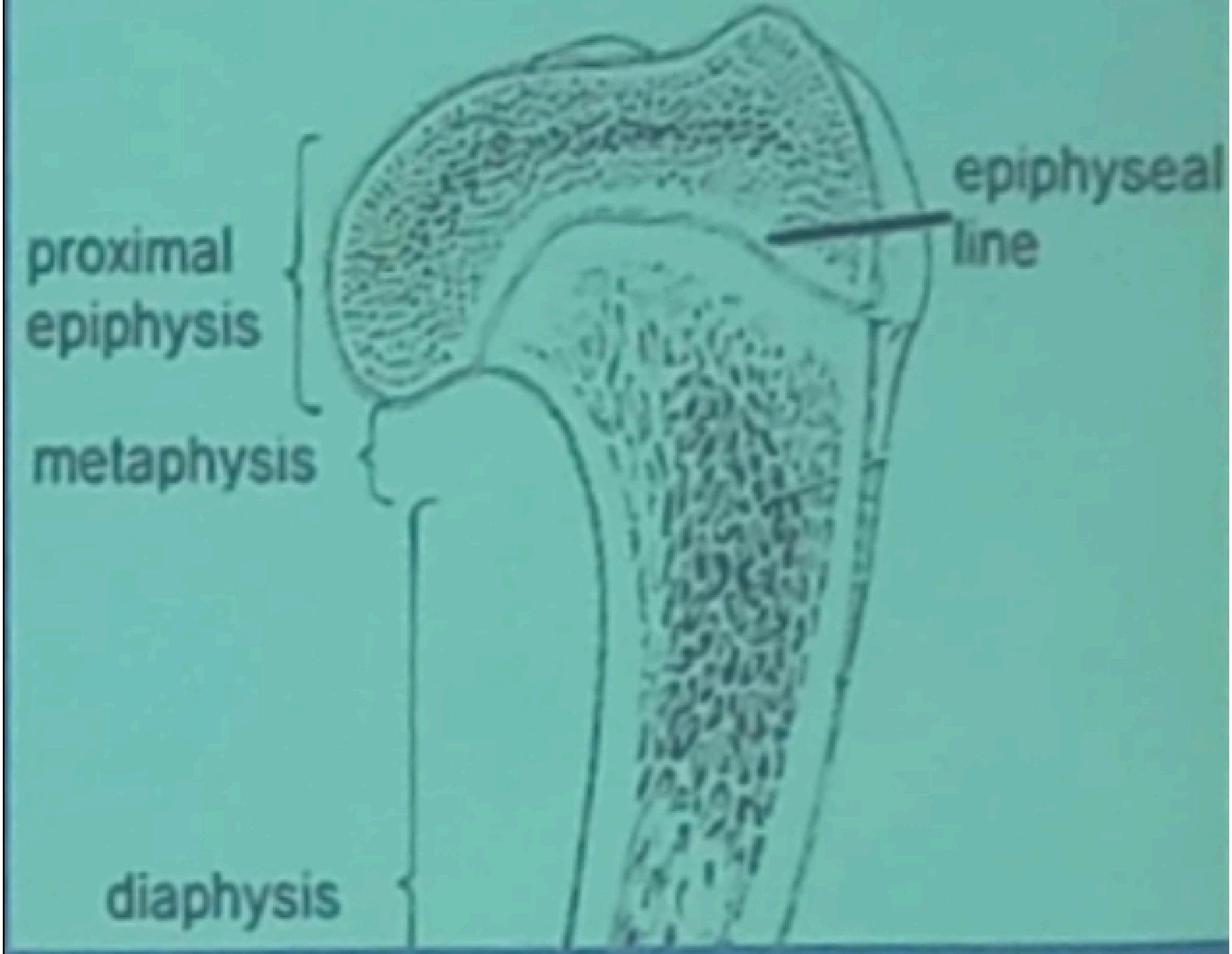
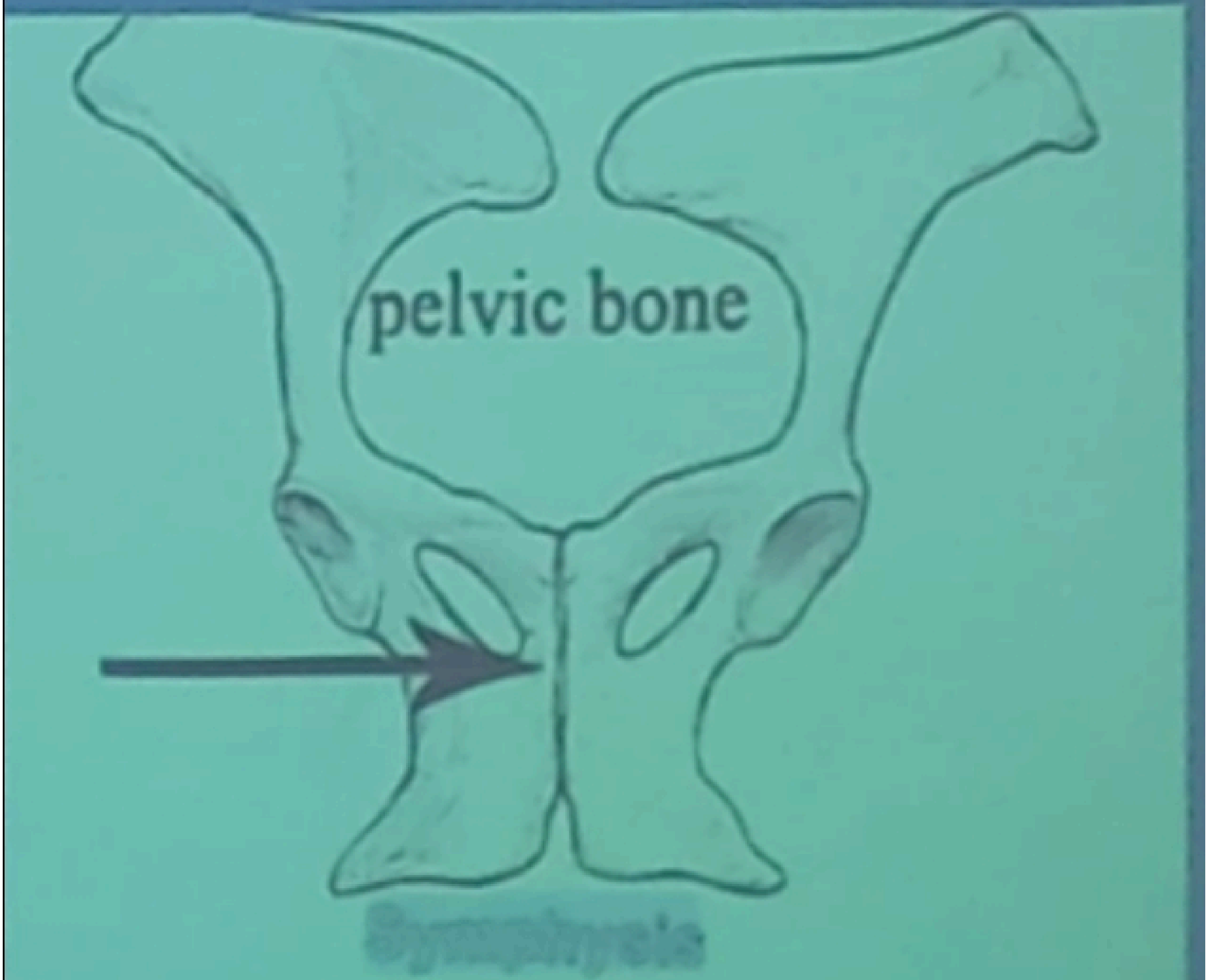
Syndesmosis: any fibrous joint outside the skull. Examples are the union of shafts of small and large metacarpal bones, fusion of the bodies of the radius and ulna and tibia and fibula in the horse.

Gomphosis: the implantation of the teeth in the alveoli by Fibrous Periodontal Ligaments






epiphyseal line
proximal epiphysis
metaphysis
diaphysis


pelvic bone

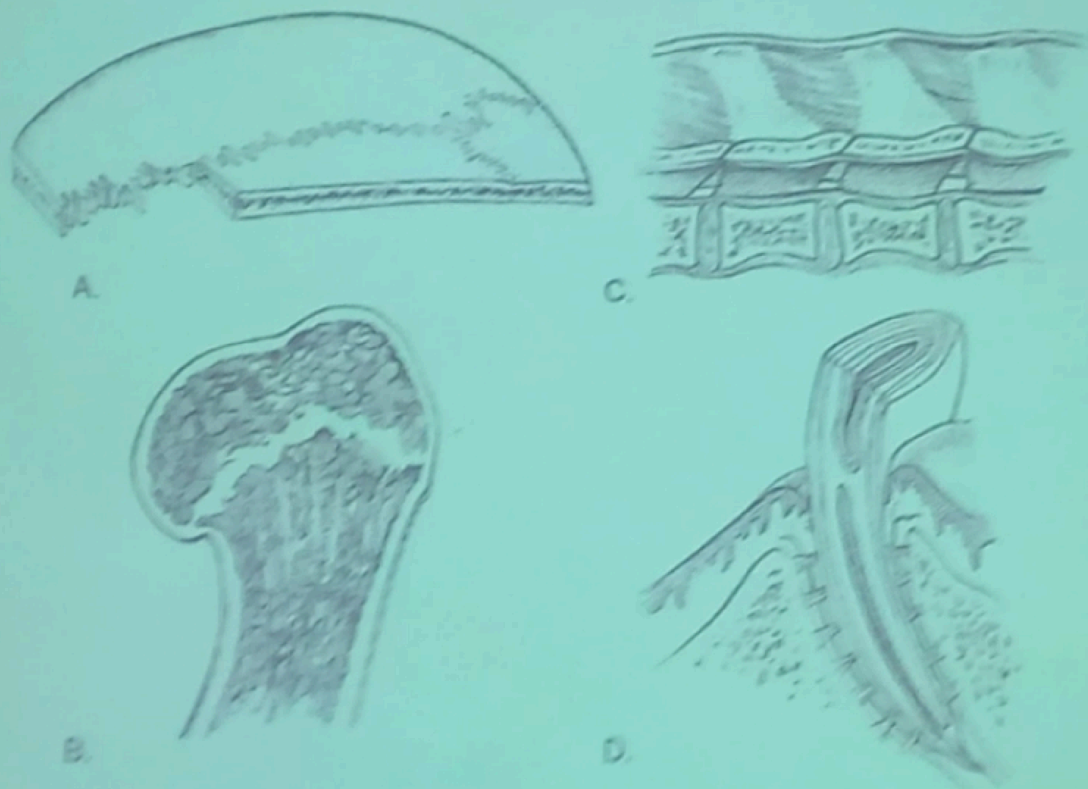
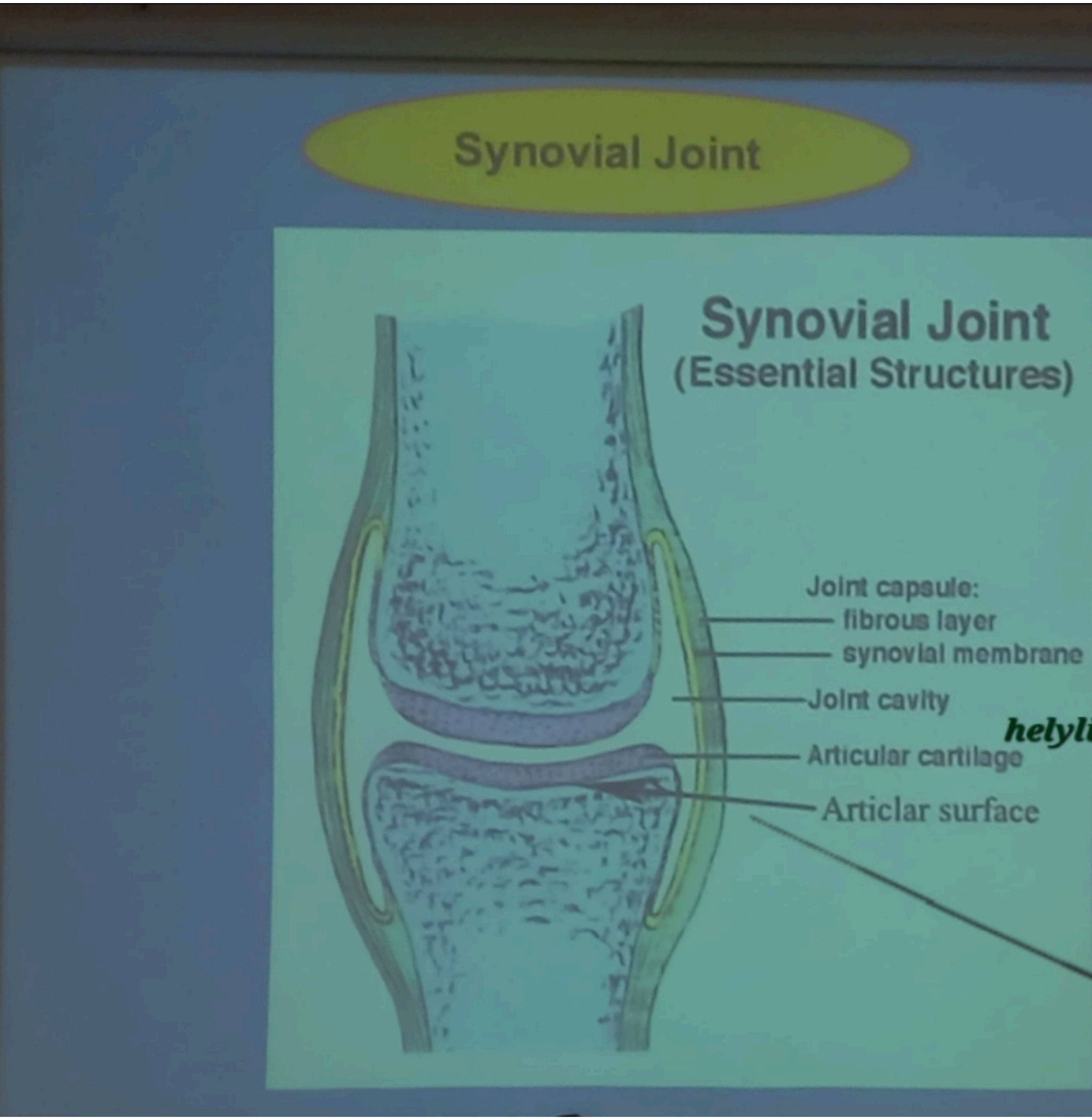
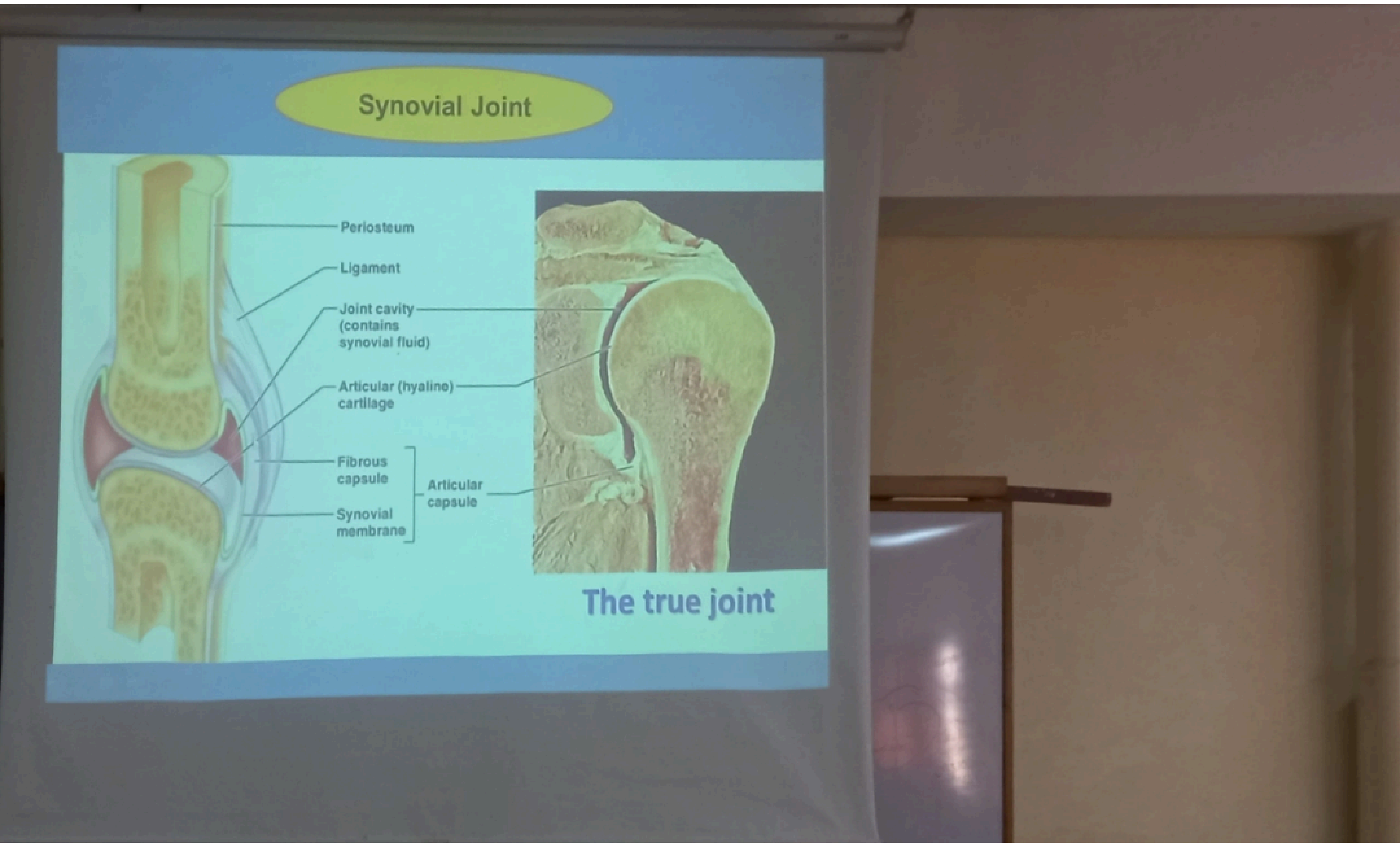


Figure 6-1. Types of non-synovial joints. A) Suture. B) Synchondrosis (growth plate). C) Symphysis (an intervertebral disk). D) Gomphosis.


Synovial Joint
Synovial Joint
(Essential Structures)
Joint capsule:
fibrous layer
synovial membrane
Joint cavity
Articular cartilage
Articlar surface
helyl



Synovial Joint
Periosteum
Ligament
Joint cavity (contains synovial fluid)
Articular (hyaline) cartilage
Fibrous capsule
Synovial membrane
Articular capsule
The true joint

Synovial Joint
diarthrodial joint

Def:

It is true movable joint are characterized by, joint capsule with its synovial membrane, joint cavity and it filled with synovial fluid

•Structure of the synovial joint

1. **Articular surface**: **smooth dens compact bone**

2. **Articular cartilage**: hyaline type ,covering over the articular surfaces of the bone. (avascular)

3. **Articular capsule (joint capsule): It is consists of two layers**:

   External (fibrous layer); its function protection the joint.

   Internal (Synovial layer): synovial membrane (contain synovial villi),secreting the synovial fluid (synovia) which lubricates the joint, which It also serves to transport nutrient material to the hyaline articular cartilage.

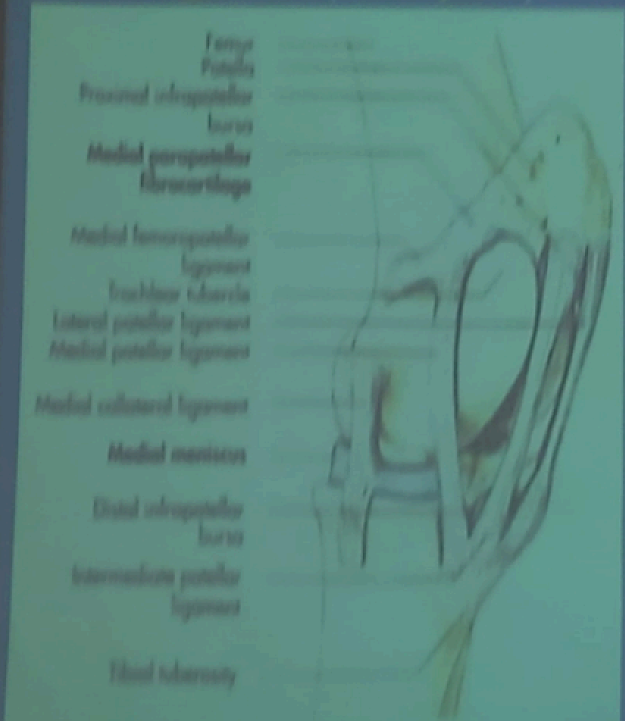
4- Ligaments

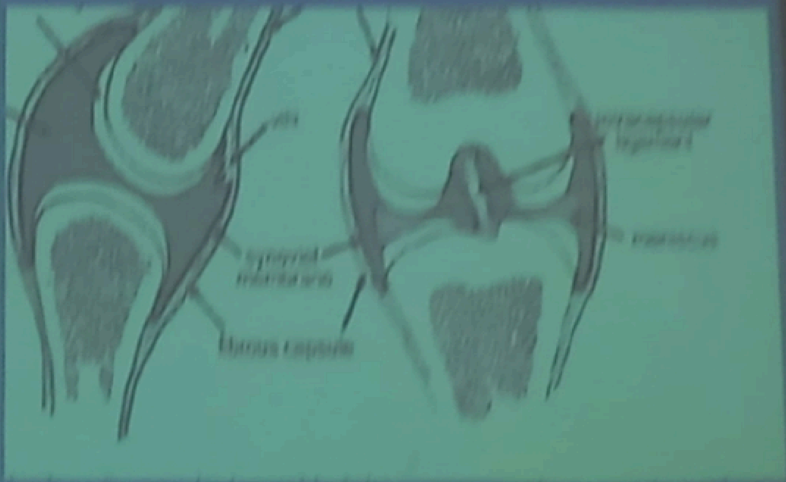
are strong bands white fibrous tissue(CT) between bones

The ligaments may be subdivided according to the position into

1 Extra capsular ligament e.g. collateral ligament

2 Intra capsular ligament situated in the joint cavity e.g. cruciate ligament (stifle joint)

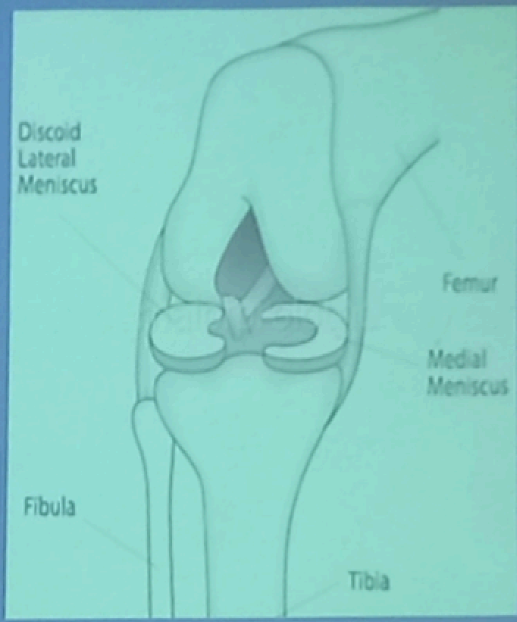


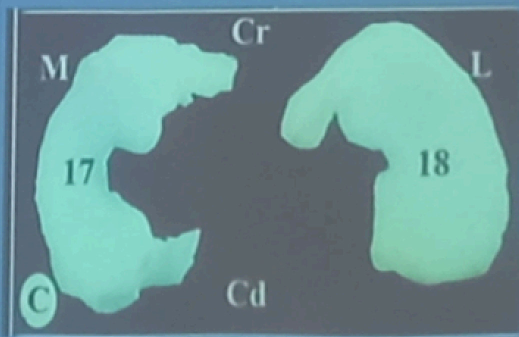


5- Articular disc and menisci:

- These are plate of fibrocartilage placed between the articular surfaces.
- Convert incongruent surface to congruent surface and diminish the concussion e.g.

Meniscu	Disc
is semilunar shape	is oval shape and divided the joint cavity into two cavities,
EX. menisci of stifle joint and	EX. articular dis in tempera mandibular joint
Stifle joint	Disc at tempromandibular joint





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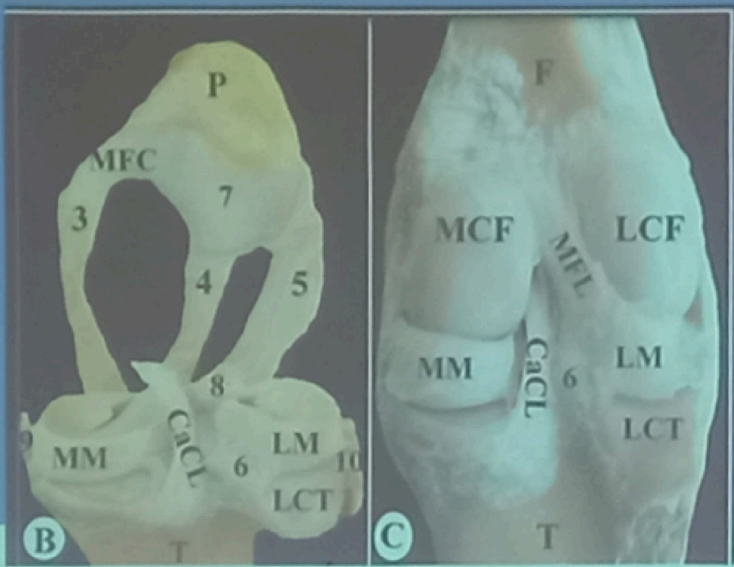
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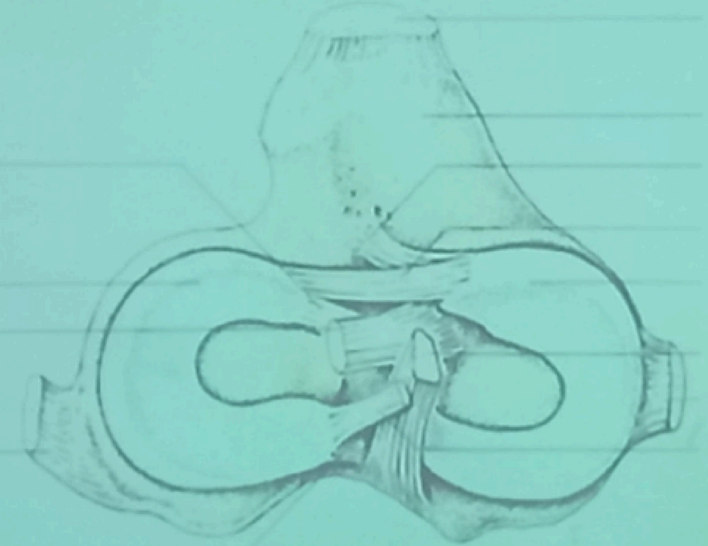
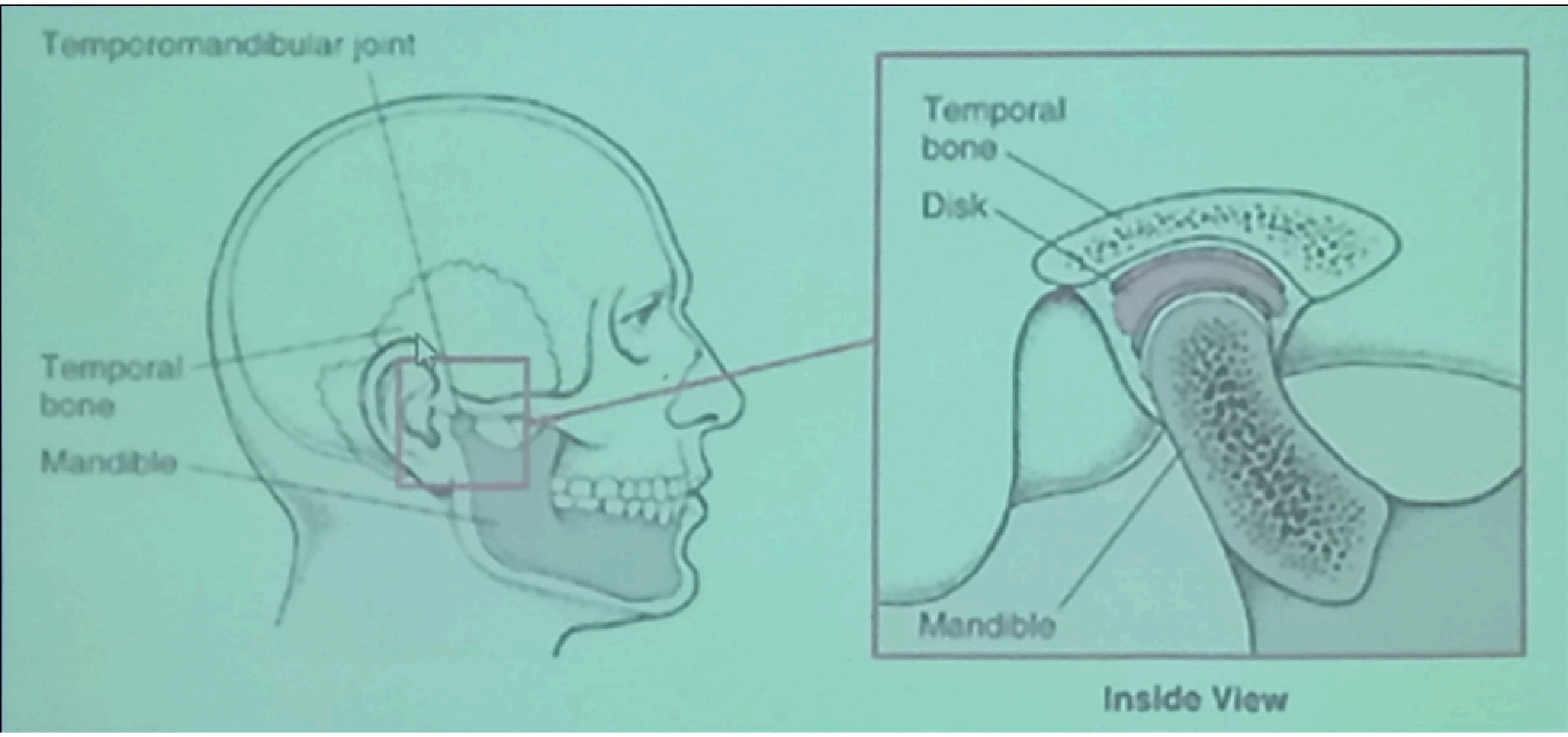
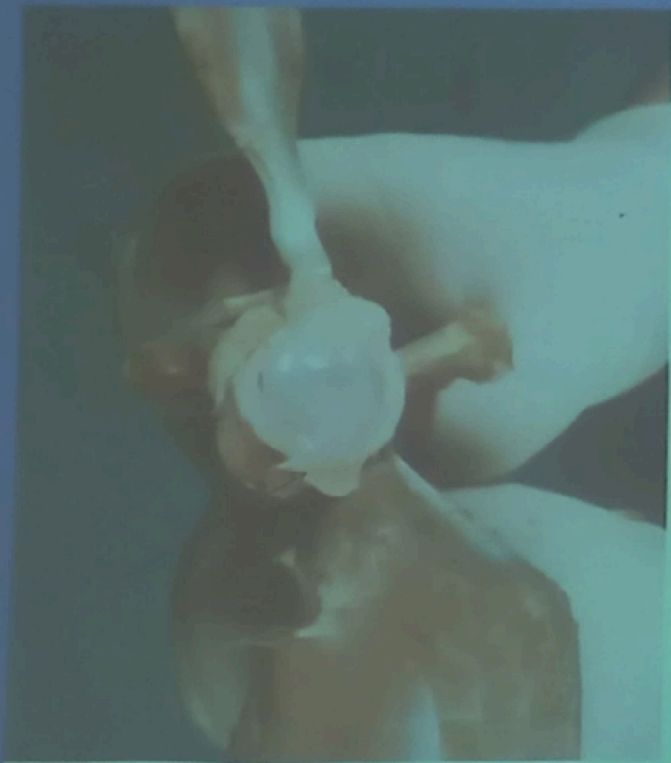

Patellar ligament
Tibial tuberosity
Cranial tibial ligament of lateral meniscus
Cranial tibial ligament of medial meniscus
Transverse ligament of stifle
Lateral meniscus
Medial meniscus
Cranial cruciate ligament
Caudal cruciate ligament
Medial collateral ligament
Lateral collateral ligament
Meniscofemoral ligament
Caudal tibial ligament of lateral meniscus


Temporomandibular joint
Temporal bone
Mandible
Temporal bone
Disk
Mandible
Inside View

6- Marginal cartilage

It is a ring of fibrocartilage which encircles the rim of the articular cavity. It enlarges the cavity and tends to prevent fracture of the margin e.g. marginal lip of acetabulum. and marginal lip of glenoid



Glenoid – scapula

Humeral head

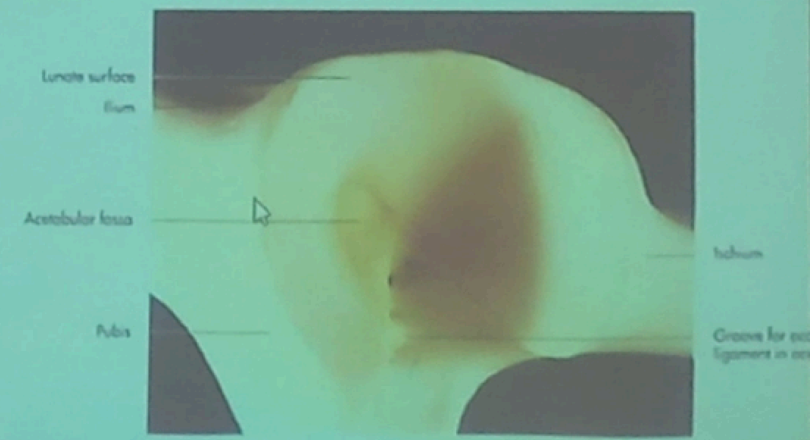
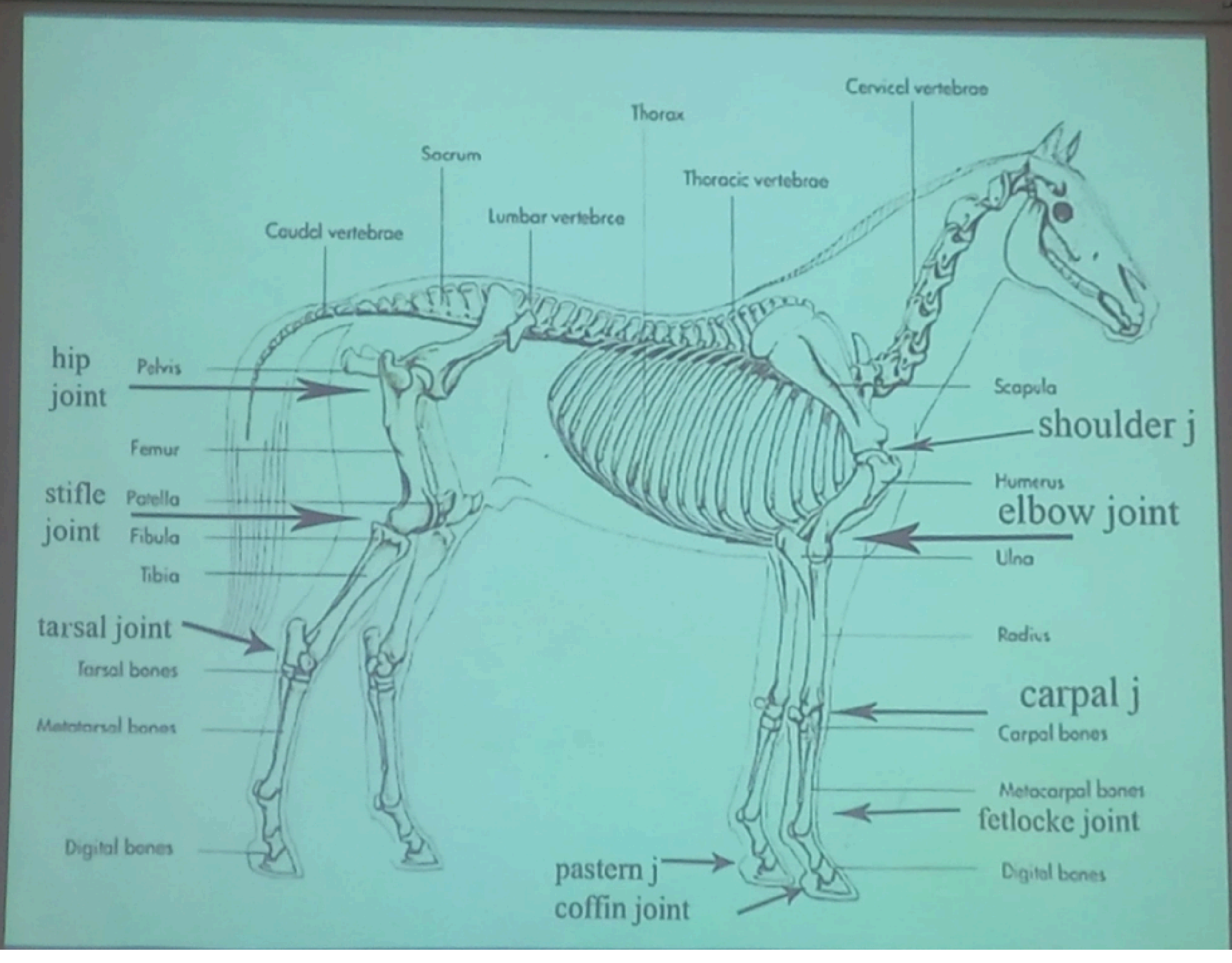


Fig. 4-12. Left acetabulum of a horse (lateral aspect).

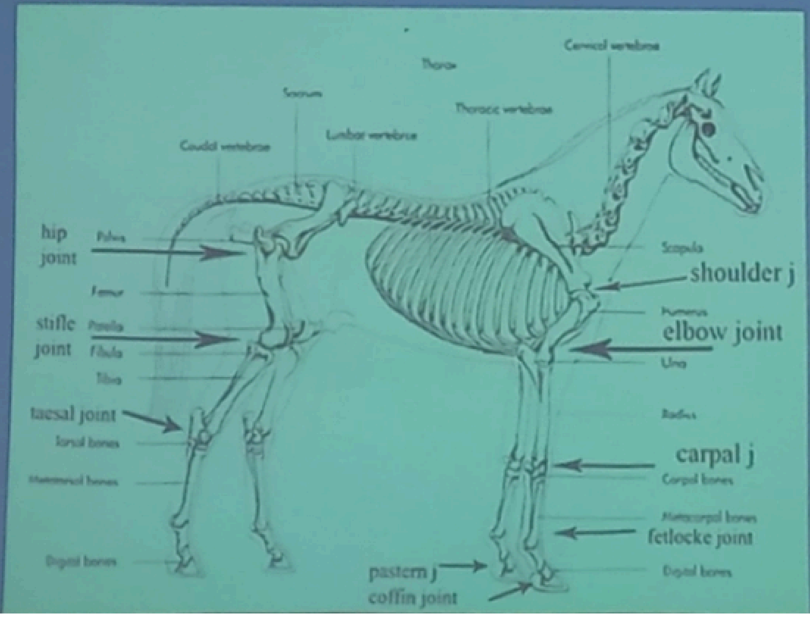


Cervical vertebrae
Thorax
Sacrum
Thoracic vertebrae
Lumbar vertebrae
Caudal vertebrae
hip joint
Pelvis
Femur
stifle joint
Patella
Fibula
Tibia
tarsal joint
Tarsal bones
Metatarsal bones
Digital bones
Scapula
shoulder j
Humerus
elbow joint
Ulna
Radius
carpal j
Carpal bones
Metacarpal bones
fetlocke joint
Digital bones
pastern j
coffin joint

Anatomical classification of the synovial joint

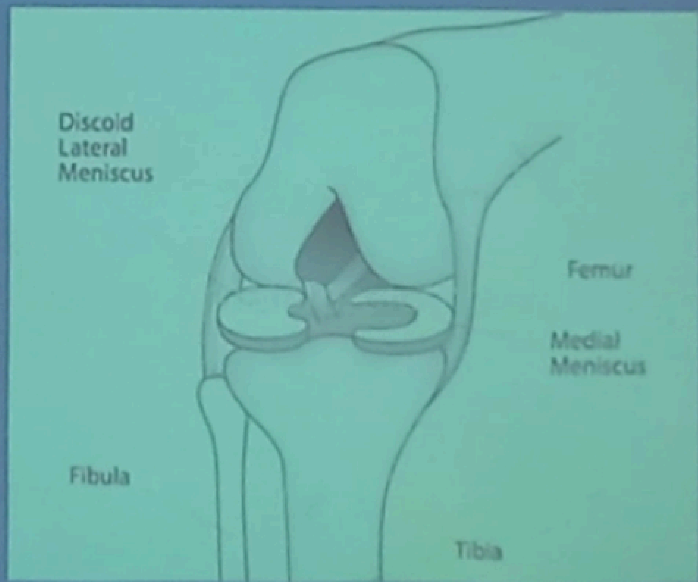
1- According to the number of bones:

- **Simple joint:** Consists of two bones
  e.g. shoulder joint and hip joint
- **Compound joint**: consists of more than two bones
  e.g. elbow joint, carpal joint and tarsal joint



2- According to the fitting of the articular surface:

- **Congruent joint**: in which the two articular surfaces are adapted to each other convex and concave articular surface e.g. shoulder joint.
- **Incongruent joint**: in which the articular surface are not adapted to each other,(convex) and convex articular surface e.g. stifle joint so that there is a menisci



Movements of synovial joints

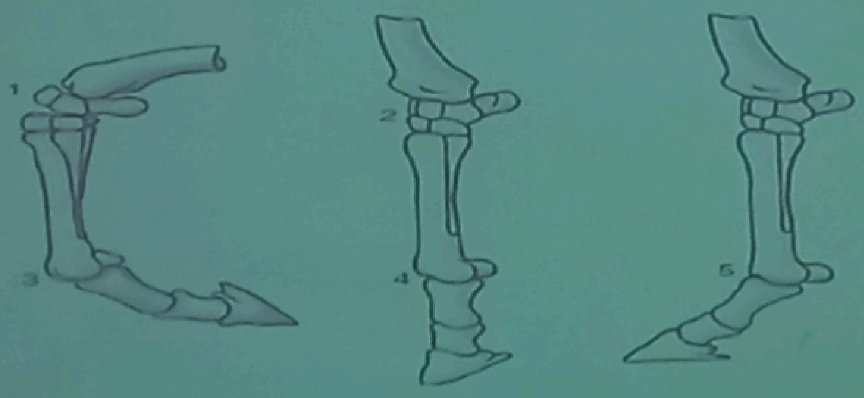
1) Gliding or sliding: when the flat surfaces glide over each other. e.g. within carpus and tarsus.

2) Angular movement:

b) **Flexion** decreases the angle between two bones.

c) **Extension** increases the angle between two bones.

d) **Hyperextension** if extension is greater than 180 degrees. E.g. fetlock joint is hyperextended in the normal standing position.



1, Flexed carpal joint; 2, extended carpal joint; 3, flexed fetlock joint;4, extended fetlock joint; 5, overextended fetlock joint.

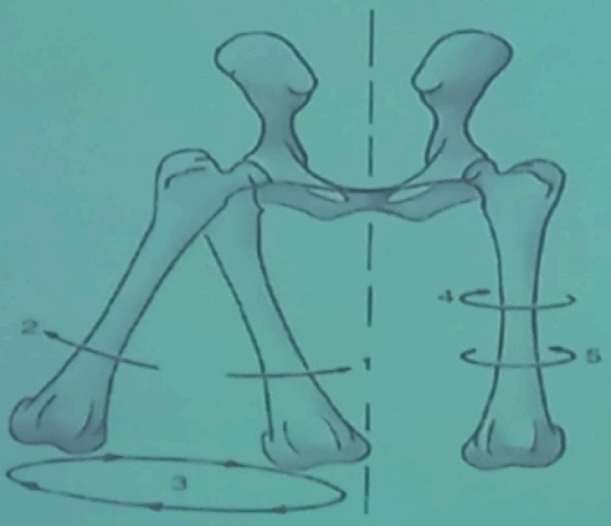
e) **Adduction** movement toward the median plane.

f) **Abduction** movement away from the median plane.

3) Rotatory movement:

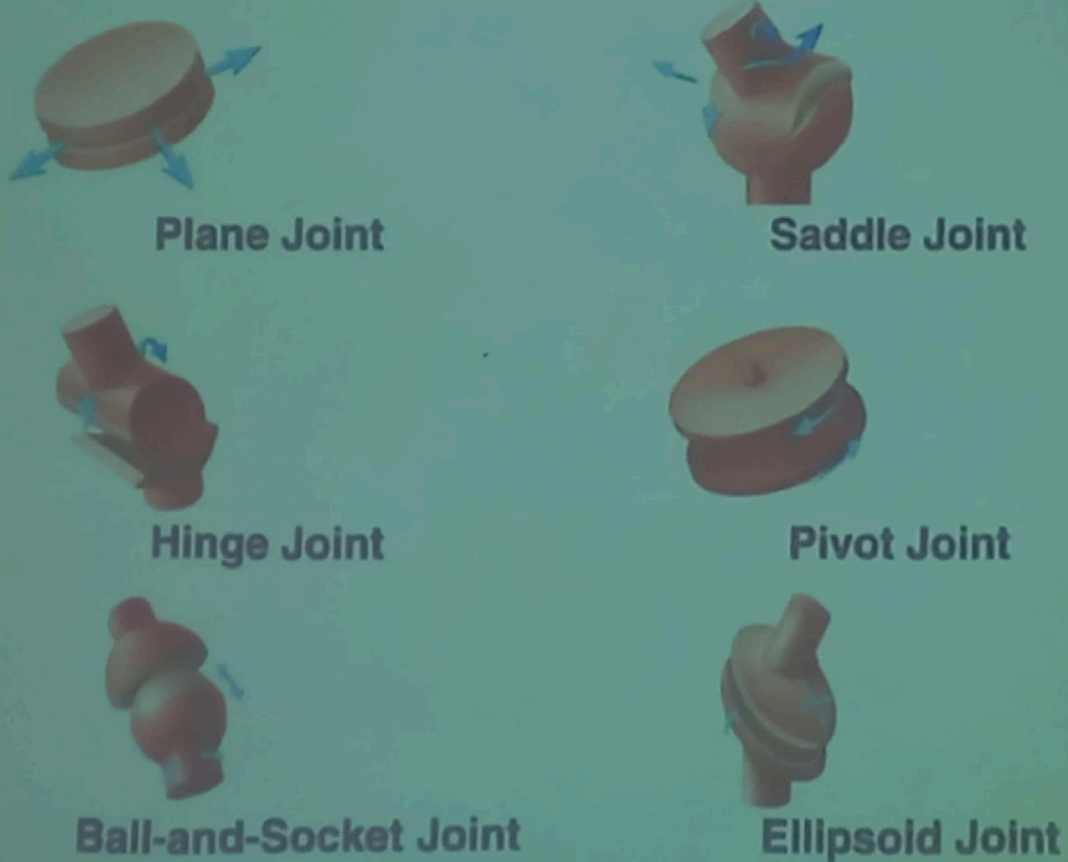
g) **Axial rotation** movement of a bone around its long axis.

h) **Circumduction (Conical rotation)** a combination of movements (flexion, abduction, extension and adduction), allowing the limb to outline a cone.



1, Adduction; 2, abduction; 3, circumduction; 4, inward rotation; 5, outward rotation.

III. Articulations- Types of Synovial Joints

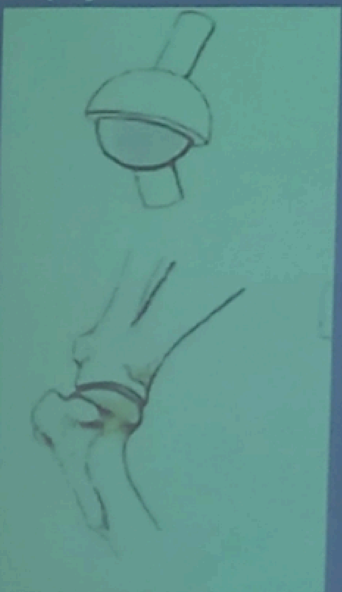


3-According to the shape of the articular surfaces:

1- Ball and socket (Multiaxial joint)

1.Spheroidal: in which the ball larger than socket e.g. shoulder joint.

2.Enarthrosis: in which the socket larger than ball e.g. hip joint.



Spheroidal joint, e.g. shoulder joint